ADITYA ASHOK SINGH

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SKILLS

Languages: JavaScript, Java
Backend: Node.js, Express.js, Prisma, Zod
Frontend: React, HTML5, CSS3, Tailwind
Databases: MongoDB, PostgreSQL, Redis
DevOps & Tools: Docker, Git, GitHub Actions, Jenkins, AWS, Linux, Terraform, Ansible
Concepts: REST APIs, JWT Authentication, RBAC, CI/CD, Data Structures & Algorithms
Soft Skills: Adaptive Learner, Problem-Solving

PROJECTS

AI-Powered Git Repository Analyzer (Containerized DevOps Automation Tool)

Dec 2025 – Present

- Developed a Dockerized Git repository analysis platform that analyzes local and remote Git repositories, extracting commit statistics, contributor distribution, code churn, hotspot files, and stale branch insights.
- Built a parameterized Jenkins pipeline that dynamically clones remote Git repositories, executes the analyzer inside Docker, and archives generated reports and visualizations.
- Designed a modular AI reporting pipeline separating prompt construction, Groq LLM integration, report generation, and chart generation using Node.js and LangChain, with graceful static-report fallback.
- Automated Markdown report generation and repository visualizations for contributor distribution and daily commit activity using Node.js, Chart.js, and Bash.
- Implemented GitHub Actions CI/CD to build and publish versioned Docker images, validate the latest image through automated analyzer execution, and upload generated reports as CI artifacts.

Tech Stack: *Bash, Git CLI, Node.js, Chart.js, LangChain, Groq API, Docker, GitHub Actions, Jenkins*

AlgoVault

Oct 2025 – Present

- Engineered a sandboxed code execution engine supporting JavaScript, TypeScript, and Python 3 with in-memory TypeScript transpilation via the Compiler API, enforcing a 3.5-second timeout and 128 MB memory cap with per-test-case verdicts and stdout/stderr capture.
- Designed a dynamic starter-code generator deriving JavaScript, TypeScript, and Python function templates purely from question titles and test-case input shapes, with zero manual maintenance per new problem.
- Architected Zod 3-validated request pipelines across all API endpoints using reusable validateRequest middleware for body, query, and params, ensuring type-safe contracts end-to-end.
- Integrated Monaco Editor with per-language code save/restore, custom input runner, and automatic switch to the Results tab on execution, alongside a zero-dependency hash-based router with filter/search state preservation on back-navigation.
- Containerized the three-service stack (PostgreSQL 16, Express API, Nginx-served React) using multi-stage Dockerfiles and Docker Compose, with health-check-gated service startup and a non-root runtime user for security.

Tech Stack: *Node.js, Express 5, TypeScript, PostgreSQL, Prisma ORM 7 (PostgreSQL Driver Adapter), Zod 3, React 19, Vite, Tailwind CSS v4, Lucide Icons, Monaco Editor, Docker, Docker Compose, Nginx, Node.js/Python subprocess execution*

TRAINING

Summer Training – Full Stack Development in React and Node

Jun 2025 – Jul 2025

Lovely Professional University

- Built a full-stack Quiz Maker platform supporting role-based authentication for 2+ user types (teachers and students).
- Created REST APIs handling real-time quiz submissions and auto-evaluation for 100+ simulated users.
- Reduced quiz evaluation time by approximately 80% through automated scoring logic.
- Enforced secure JWT-based authentication with protected routes.

CERTIFICATIONS

- Cloud Computing – IIT Kharagpur, 8-week course (NPTEL) | Oct 2025
- Introduction to MongoDB (MongoDB) | Jun 2025

EDUCATION

Lovely Professional University | B.Tech CSE, CGPA: 8.11 | Phagwara, Punjab | Aug 2023 · Present

St. Basil·s School | Intermediate: 82% | Khirighati, Uttar Pradesh | Jul 2020 · May 2022

St. Basil·s School | Matriculation: 90.6% | Khirighati, Uttar Pradesh | Apr 2018 · Mar 2020